Using statistical learning models with household survey microdata

Applications to energy demand, energy poverty, and environmental taxation

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EAERE Summer School in Resource and Environmental Economics 2026

Recent applications of AI in Environmental and Resource Economics

Session 0

Introduction

Getting to know each other

A few words about me

- **Postdoctoral researcher** at the Basque Centre for Climate Change.
- **PhD in Economics** from the University of the Basque Country.
- **Research interest:** distributional impacts of energy and climate policies.
- **Applied expertise**: use of SL models in R with household survey data.

A few words about you

- Briefly introduce yourself.
- What are your research interests?
- Self-assessment: rate your experience as low, medium, or high:
  - Statistical learning models.
  - Programming in R (or other languages)
  - Household survey microdata.

Learning objectives

What students should be able to do by the end of the course

- Understand the landscape of the main public household surveys used in energy and environmental research.
- Learn how to build policy-relevant indicators from these microdata.
- Apply statistical-learning workflows in R using household survey microdata.
- Apply classification, regression, and hurdle (two-steps) models.
- Compute predictions to help answer policy-impact and simulation questions.

Key message *The course is not a general introduction to machine learning. The focus is applying statistical-learning tools to household survey microdata to answer environmental-economics questions.*

Course structure

Session 0: Introduction

- Course objectives, workflow and structure of the practical sessions.
- Overview of the progression: survey microdata → indicators → models → interpretation → policy questions

Session 1: Household surveys and main indicators

- Public survey sources: HBS, EU-SILC, and other surveys.
- Main household-level indicators for energy poverty and energy demand and distributional analysis.
- Hands-on 1: Starting with household consumption microdata of Spain

Session 2: Modelling approaches and Statistical learning models

- Introduction to modelling approaches using household survey microdata.
- Models: training, validation and comparison.
- Hands-on 2: Modelling energy demand and energy poverty

Session 3: Applications and policy implications

- Applications in the literature.
- A real policy-relevant application: the case of the first Atlas of energy poverty of Spain.
- Open discussion and brainstorming

Note *Each hands-on includes unsolved and solved R scripts*

Course materials repository

Where to find the course materials and how they are organised

Repository

https://github.com/bc3LC/lecture_household-surveys-ml

Software requirements

Download and install the required software before the practical sessions

R

Download from:
https://www.r-project.org/

RStudio

Download from:
https://www.rstudio.com/

Recommended *Install packages from the scripts in advance, then run the first lines to check that the project root and data folders are detected correctly.*